



Industrial HMI Display

EPC-ESP32P4-070



PN: CS10600-ESP32P4-070E

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Contents

EPC-ESP32P4-070	3
1. Product Overview	7
2. Ordering Options	8
3. Hardware Features	9
3.1. Optional Features	11
4. Power Input	12
5. Connectivity	14
5.1. RS232/RS485/CAN/Direct GPIO	14
5.2. Isolated GPIO	16
5.3. USB	19
5.4. LAN	20
5.5. 4G/LTE	21
5.6. MicroSD Card	22
5.7. Audio	23
5.7.1. 3.5 mm Audio In/Out	23
5.7.2. Internal Speaker	23
5.7.3. Buzzer	23
6. Button and Status LED	24
6.1. Status LED	24
6.2. Button	25
7. Mounting Procedure	26
8. Mechanical Specifications	27
9. Disclaimer	28
10. Technical Support	28

EPC-ESP32P4-070

Front View



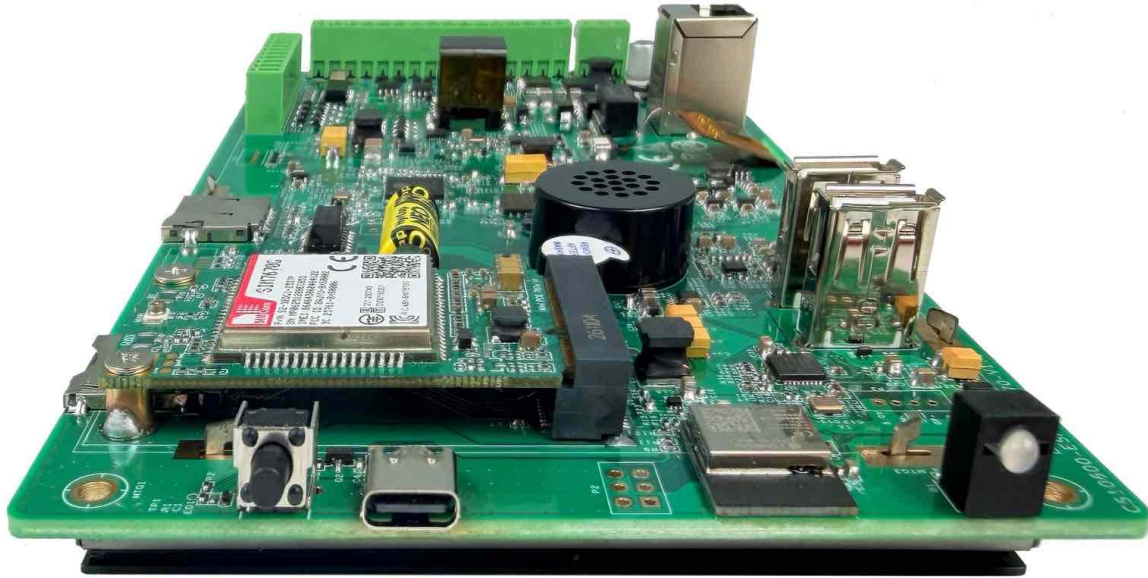
Rear View



Side View 1



Side View 2



Product Overview

The EPC-ESP32P4-070 Embedded HMI Display (PN: CS10600-ESP32P4-070E) is an embedded programmable capacitive touch HMI display featuring an ESP32 P4 processor. The product features an embedded mounting design, making it suitable for integration into existing equipment.

It features a 7" IPS display with a maximum brightness of 400 cd/m². The touch screen is a responsive capacitive screen and supports multi-touch.

The product is powered by an ESP32-P4 processor, making it suitable for applications such as:

- Smart Home
- Industrial Automation
- Healthcare
- Consumer Electronics
- Smart Agriculture
- Retail Self-Service Terminals (POS, Vending Machines)
- Service Robots
- Multimedia Player
- High-Speed USB Host and Device
- Smart Voice Interaction Terminal
- Edge Vision AI Processor
- HMI Control Panel

The product can be programmed with ESP-IDF, ESPHome and other ESP32 ecosystem tools and frameworks that support the ESP32-P4.

Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the [Hardware Features](#) section provides information about the default options bundled with the product.



Note

You can order [EPC-ESP32P4-070](#) from the official [Chipsee Store](#) or from your nearest distributor.

Hardware Features

The EPC-ESP32P4-070 Embedded HMI Display offers a broad range of performance and connectivity features for flexible integration. Some of the key features are listed below.

EPC-ESP32P4-070	
CPU	Dual-core 32-bit RISC-V, up to 400 MHz; single-core 32-bit LP RISC-V, up to 40 MHz
Storage	32MB external SPI Flash
RAM	768 KB HP L2MEM + 32 KB LP SRAM + 8 KB SPM; 32MB in-package PSRAM
Display	7" IPS LCD, 1024 x 600 px, brightness 400 cd/m ²
Touch	5-point capacitive touch with 1.0mm tempered glass
LAN	1 x 100Mbps LAN
PoE	Power over Ethernet 802.3af/at, optional, not mounted by default
WiFi/BT	WiFi 6, Bluetooth LE 5 (optional, ESP32-C6 is not mounted by default)
4G/LTE	Internal Cat 1 4G/LTE module, not mounted by default
Audio	3.5mm audio in/out interface; internal dual channel speaker
Buzzer	Onboard buzzer, driven by I2C expanded IO
GPIO	8 channels isolated I/O, 4 x input and 4 x output. 4 channels of non-isolated P4 direct out GPIO
RS232	2 x RS232 (default); up to 4 x RS232 (optional , by configuring RS485 to RS232)
RS485	2 x RS485 (default); 2 x RS485 can be configured to 2 x RS232 (optional)
CAN	1 x CAN (default)
USB	2 x USB 2.0 Type-A high-speed host ports; 1 x USB 1.1 Type-C port for firmware flashing and debugging; the type-C port supports powering the device while programming
Micro SD card socket	1 x TF (micro SD) card slot
RTC	High-accuracy RTC with supercapacitor, can work 1 week after power off (default). High accuracy RTC with lithium coin battery, can work 3 years after power off (optional).
Power Input	9V to 32V DC
Current	260mA max at 15V, 200mA typical at 15V
Power Consumption	4W max, 3W typical
Operating Temp.	From -10°C to +60°C
Dimensions	190 x 107.8 x 27.7mm

EPC-ESP32P4-070	
Weight	400g
Mounting	Embedded
Certification	CE, RoHS

Table 503 Key Features

Optional Features

- WiFi/Bluetooth: If Wi-Fi/Bluetooth connectivity is required, an ESP32-C6 module can be installed at the factory. The ESP32-C6 is not populated by default.
- 4G/LTE: 4G/LTE (Cat 1) module is supported on this product, it's not mounted by default, if you need the interface or the 4G/LTE module you can contact us when you place an order.

Warning

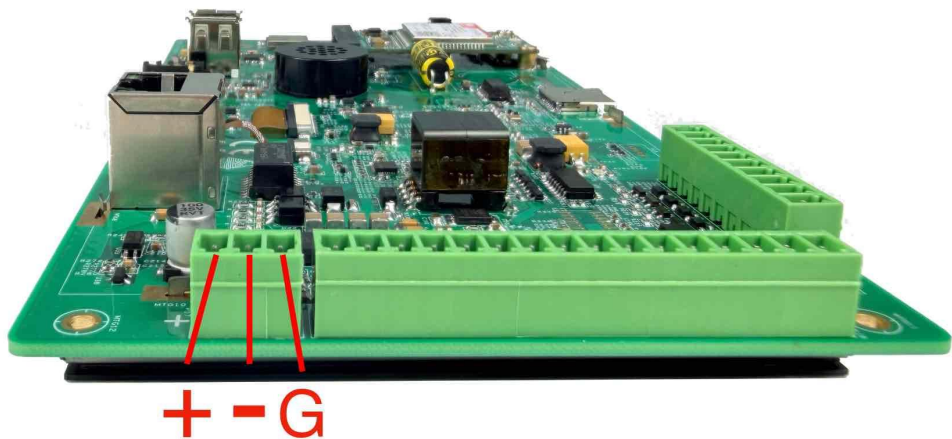
Installation, repair, and maintenance tasks should be performed by trained personnel only. Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

Power Input

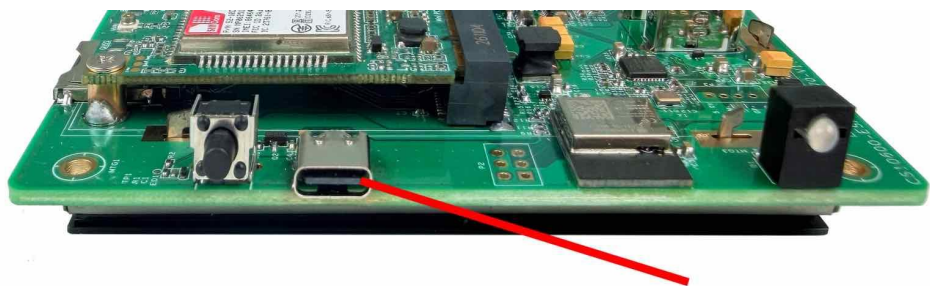
The EPC-ESP32P4-070 Embedded HMI Display can be powered by an input voltage of 9V to 32V DC.

The total power consumption is about 4W max, 3W typical, depending on the load and brightness.

The power input is **3-pin 3.81mm screw terminal**, you can also use the USB-C port to power the product while programming.



Power Connector



USB Power Input

Note that the “+” sign represents the positive power input, the “-” terminal is shorted to the ground.

Power Input Definition		
Pin Number	Definition	Description
Pin 1	Positive Input	DC Power Positive Terminal
Pin 2	Negative Input	DC Power Negative Terminal
Pin 3	Ground	Power System Ground

Table 504 Power Connector

 Note

The system ground “**G**” is connected to power negative “-” on board.

The product supports **optional** PoE (Power over Ethernet), it's **not** mounted by default.

Connectivity

There are many connectivity options available on the EPC-ESP32P4-070.

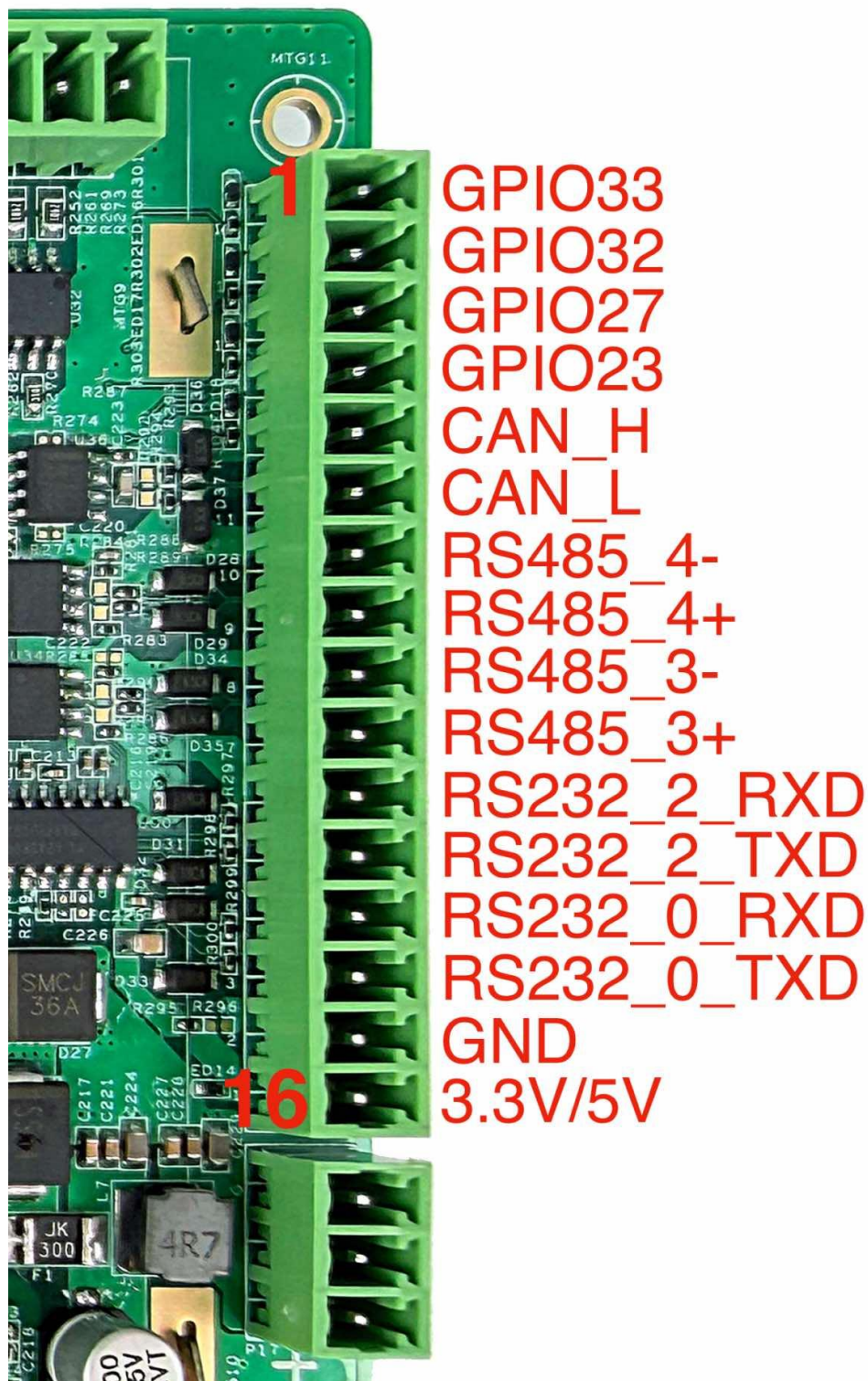
RS232/RS485/CAN/Direct GPIO

The product provides the following serial and CAN interfaces:

- 2 x RS232 **(default)**; up to 4 x RS232 **(optional)**, by configuring RS485 to RS232)
- 2 x RS485 **(default)**; 2 x RS485 can be configured to 2 x RS232 **(optional)**
- 1 x CAN **(default)**

RS232 / RS485 / CAN / Direct GPIO Pin Definition			
Pin Number	Definition	Description	ESP32 P4 Pin
Pin 1	GPIO33	Direct ESP32-P4 GPIO	GPIO33
Pin 2	GPIO32	Direct ESP32-P4 GPIO	GPIO32
Pin 3	GPIO27	Direct ESP32-P4 GPIO	GPIO27
Pin 4	GPIO23	Direct ESP32-P4 GPIO	GPIO23
Pin 5	CAN_H	CAN high signal	GPIO22
Pin 6	CAN_L	CAN low signal	GPIO21
Pin 7	RS485_4-	RS485 -(B) signal	GPIO47
Pin 8	RS485_4+	RS485 +(A) signal	GPIO48
Pin 9	RS485_3-	RS485 -(B) signal	GPIO4
Pin 10	RS485_3+	RS485 +(A) signal	GPIO5
Pin 11	RS232_2_RXD	RS232 RXD signal	GPIO2
Pin 12	RS232_2_TXD	RS232 TXD signal	GPIO3
Pin 13	RS232_0_RXD	RS232 RXD signal	GPIO38
Pin 14	RS232_0_TXD	RS232 TXD signal	GPIO37
Pin 15	GND	System ground	GND
Pin 16	3.3V/5V	Power output	3.3V/5V

Table 505 RS232/RS485/CAN Direct GPIO Connector



RS232/RS485/CAN/Direct GPIO Connector

⚠ Attention

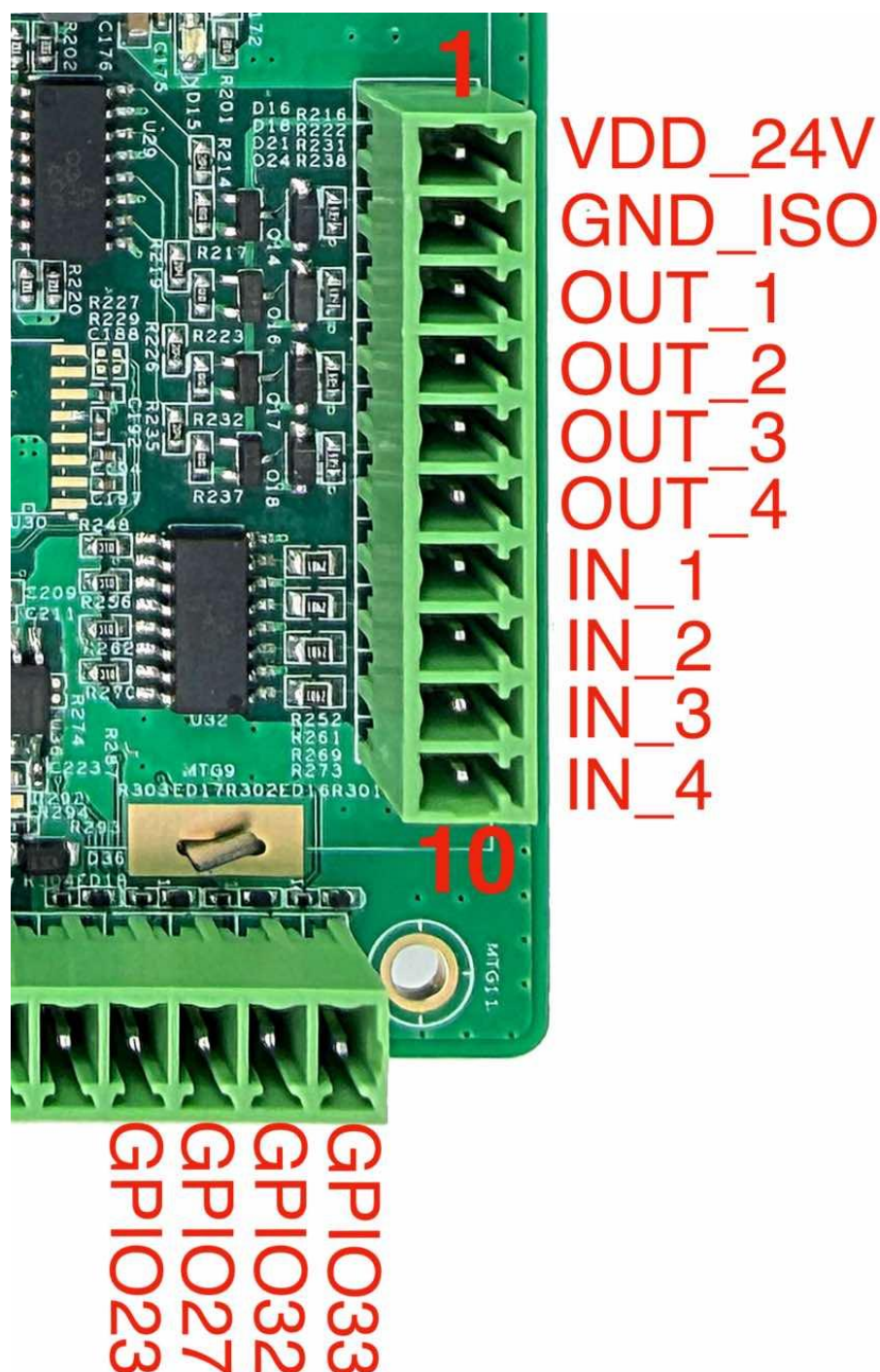
1. Pins 1 to 4 are direct ESP32-P4 GPIO signals.
2. Pin 16 is 5V by default, if you need 3.3V, you can **Contact us** before shipping.
3. RS485_3, RS485_4 automatically control input/output direction. They do not require software control.
4. The 120 Ω termination resistor for the RS485 is **not mounted** by default.
5. If you need a different configuration, you can **Contact us** before shipping, we can make a customized version of these pins for you.

Isolated GPIO

The EPC-ESP32P4-070 Embedded HMI Display has 8 optically isolated I/O channels, as shown in the figure below. These are controlled through an I2C GPIO expander at address 0x21.

Isolated GPIO Connector Pin Definition		
Pin Number	Definition	Description
Pin 1	VDD_24V (VDD_ISO)	Isolated Power +5V ~ +24V Input (Should not exceed 24V)
Pin 2	GND_ISO	Isolated Ground
Pin 3	OUT1	Isolated Output 1
Pin 4	OUT2	Isolated Output 2
Pin 5	OUT3	Isolated Output 3
Pin 6	OUT4	Isolated Output 4
Pin 7	IN1	Isolated Input 1
Pin 8	IN2	Isolated Input 2
Pin 9	IN3	Isolated Input 3
Pin 10	IN4	Isolated Input 4

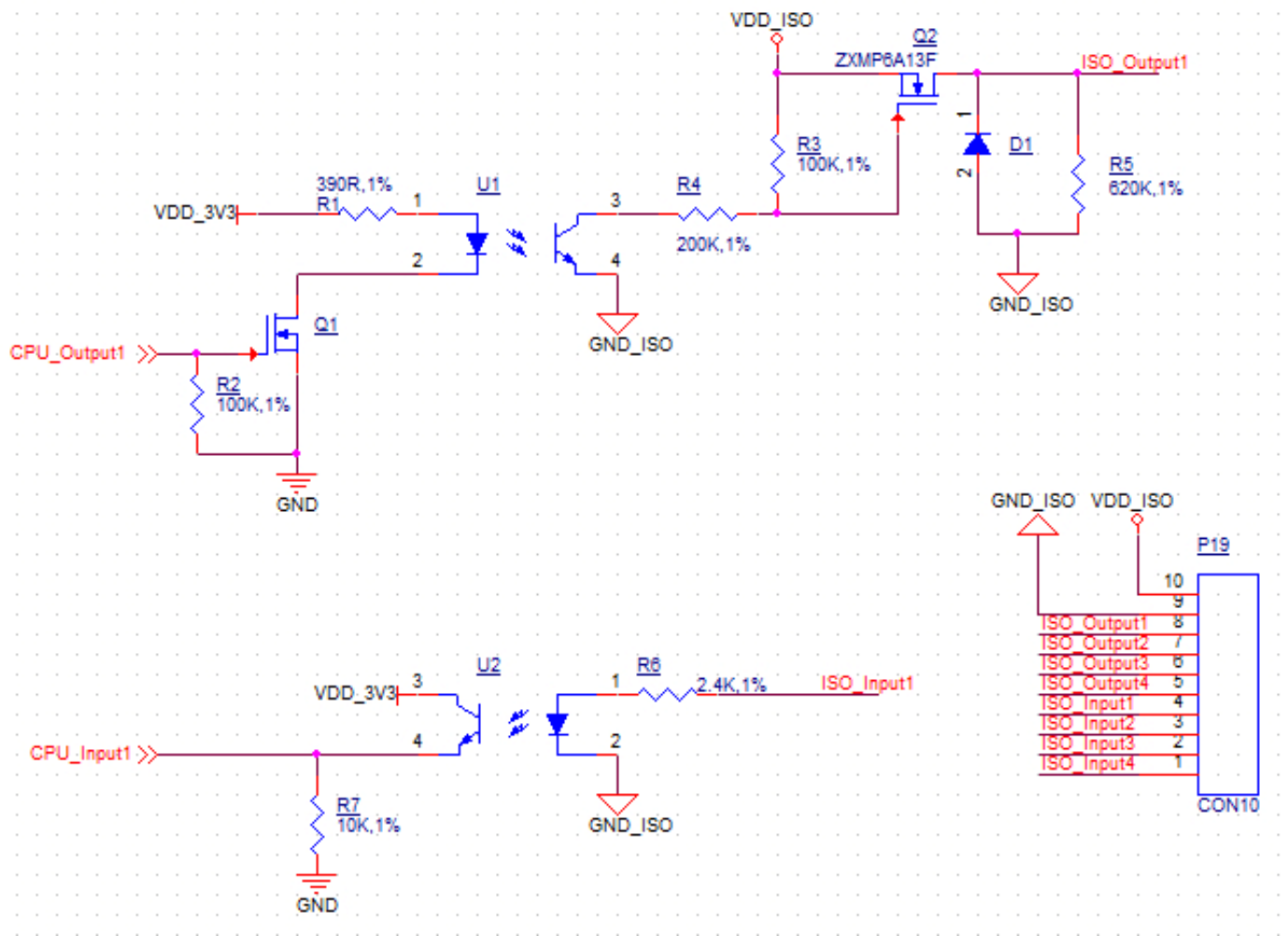
Table 506 GPIO Connector Pin-out



Optical Isolated GPIO Connector

Attention

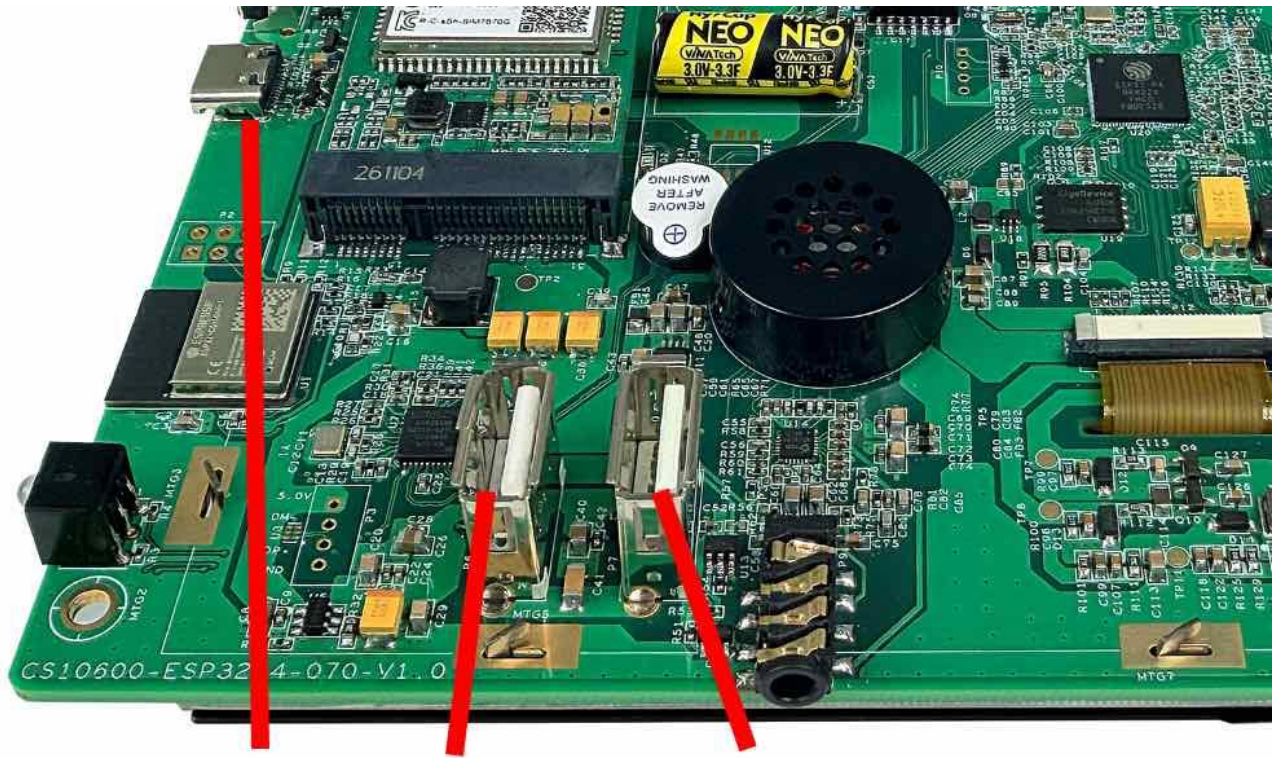
1. In order to use the Isolated Output, you need to add an external Isolated Power to the VDD_ISO and GND_ISO (between 5V and 24V). The power voltage should not exceed 24V.
2. The output current can reach 500mA for every channel, but it also depends on the isolated source that supplies the power.
3. In order to use the Isolated Input, you need to add a signal to the IN_X and GND_ISO. A 2.4KΩ resistor, as R6, has been added to limit the input current, as shown in the figure below. This resistor should work well for the 5-24V input signal. If your input signal is less than 5V, please change this input resistor. The reduced schematic is for reference purposes, if you need the precise resistor schematic, please contact us.



Isolated GPIO reduced schematic

USB

There are 2 x **USB2.0 Type-A HOST connectors** and 1 x **Type-C connector** onboard, as shown in the image below.



USB2.0 Host and Type-C Ports

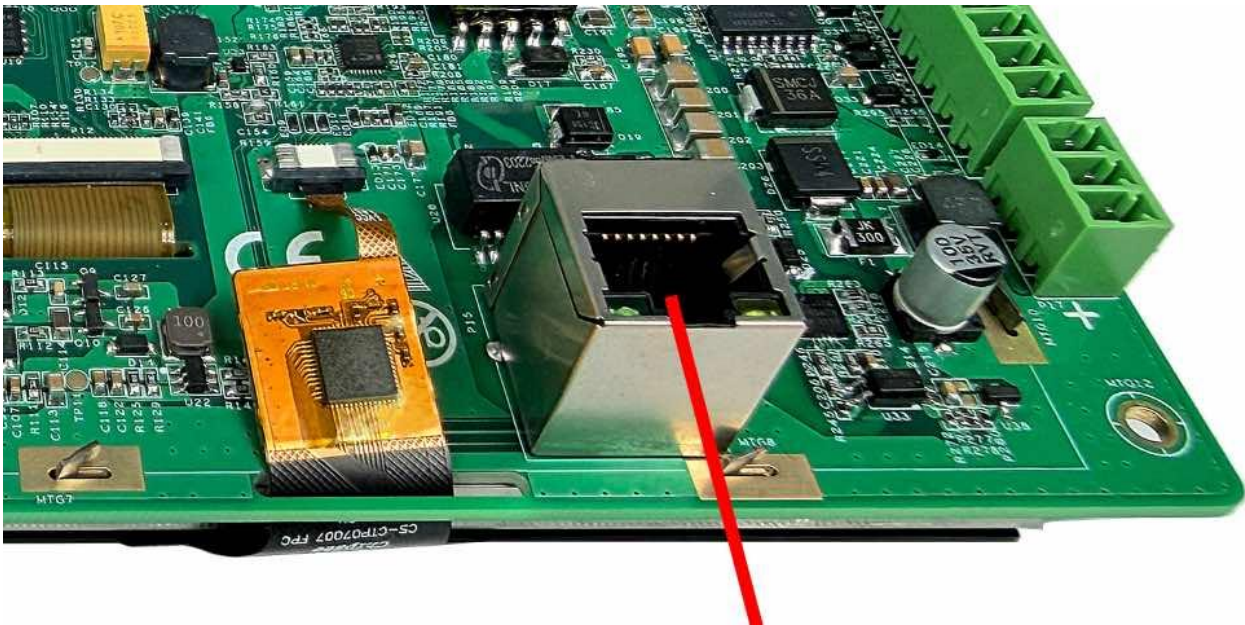
Note

- USB peripherals such as high-speed flash drives can be connected to the USB 2.0 host ports.
- Full/Low speed USB devices such as mice and keyboards **cannot** connect due to **the limitation of USB Hub Host software stack**, Espressif managed software stack doesn't support Transaction Translator layer yet.
- The USB Type-C port is used for debugging and firmware flashing and supports powering the product.

LAN

LAN (RJ45) connector provides Ethernet connectivity over standardized Ethernet cable. The integrated Ethernet interface supports up to 100 Mbps data throughput.

The product supports **optional** PoE (Power over Ethernet), it's **not** mounted by default.



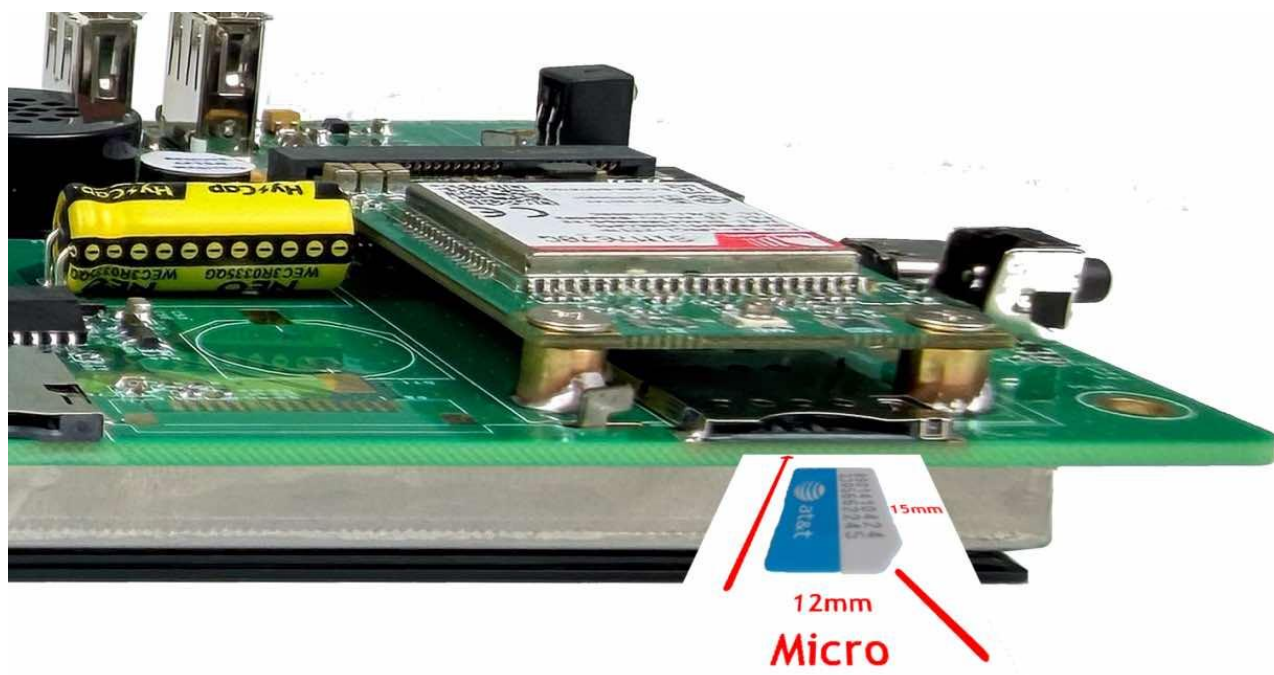
RJ45 LAN Connector

4G/LTE

4G/LTE Cat 1 module is **optional**. If you place an order with 4G module included, we will also add a SIM card holder and a 4G/LTE antenna to enable 4G/LTE connectivity.

The SIM card slot does **NOT** support hot-swapping. **Power off** the device before inserting or removing the SIM card.

The Cat 1 module doesn't support GPS.



SIM Card Direction

Attention

The product does not come shipped with the 4G/LTE module by default. If you need to use 4G/LTE, you can contact us when placing an order. We can install the necessary hardware for you.

MicroSD Card

The EPC-ESP32P4-070 Embedded HMI Display features 1 x **TF Card (micro SD) slot** (note the direction of the micro SD card). The product does not come shipped with the micro SD card by default.

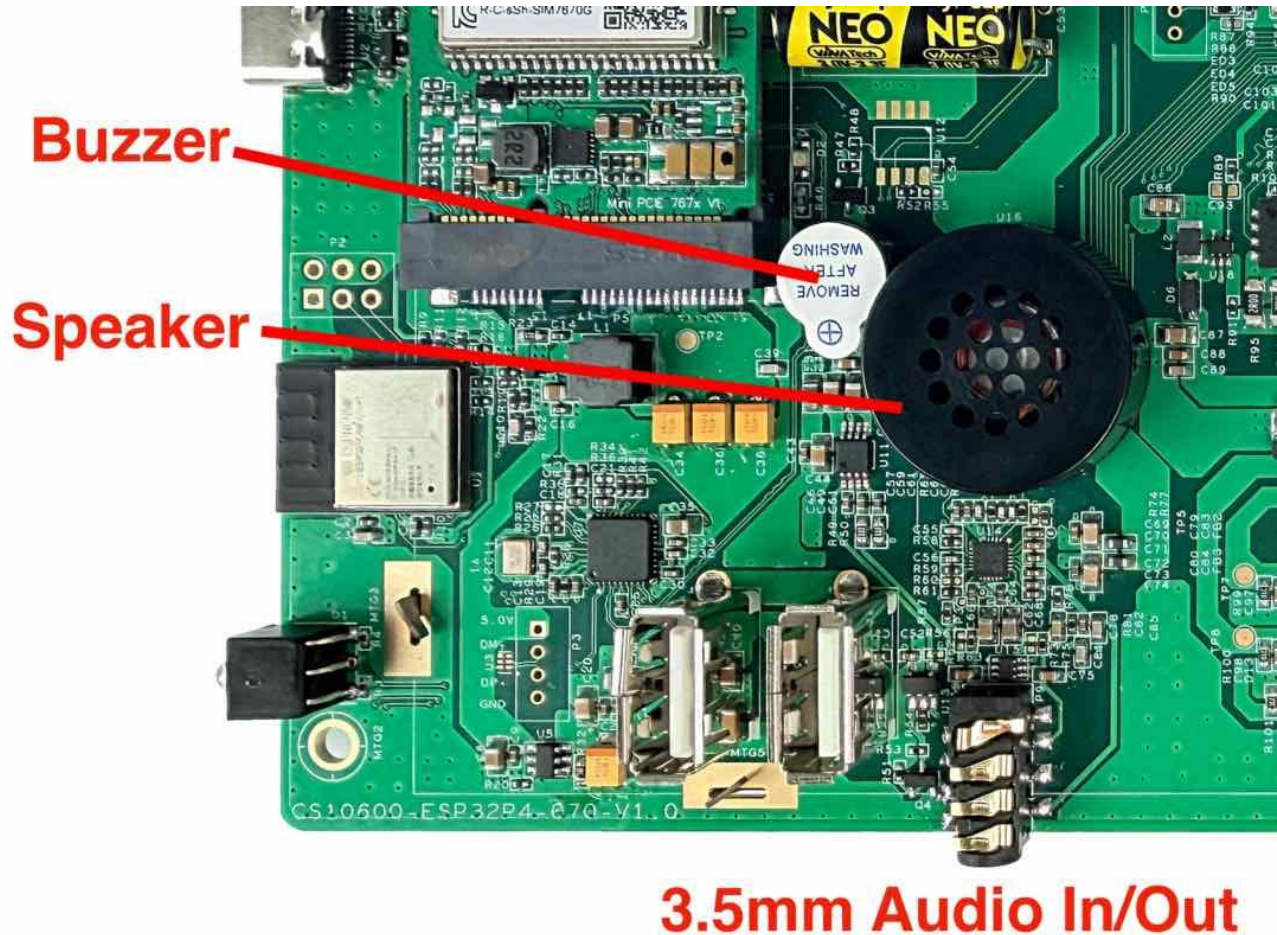


Micro SD Card Slot

Audio

3.5 mm Audio In/Out

The product features 3.5mm audio connector as shown in the image below.



Audio Connector

Internal Speaker

The product also has an internal dual channel 2W speaker, meaning you can play music without an earphone, the sound can be played from the product directly.

Buzzer

The product has an internal buzzer, suitable for playing beep sounds for alarms.

Button and Status LED

Status LED

This product has a status LED. The LED turns RED after power on and GREEN when the system is booted.



Status LED

Button

There is a PROG button for flashing firmware.



Button

Mounting Procedure

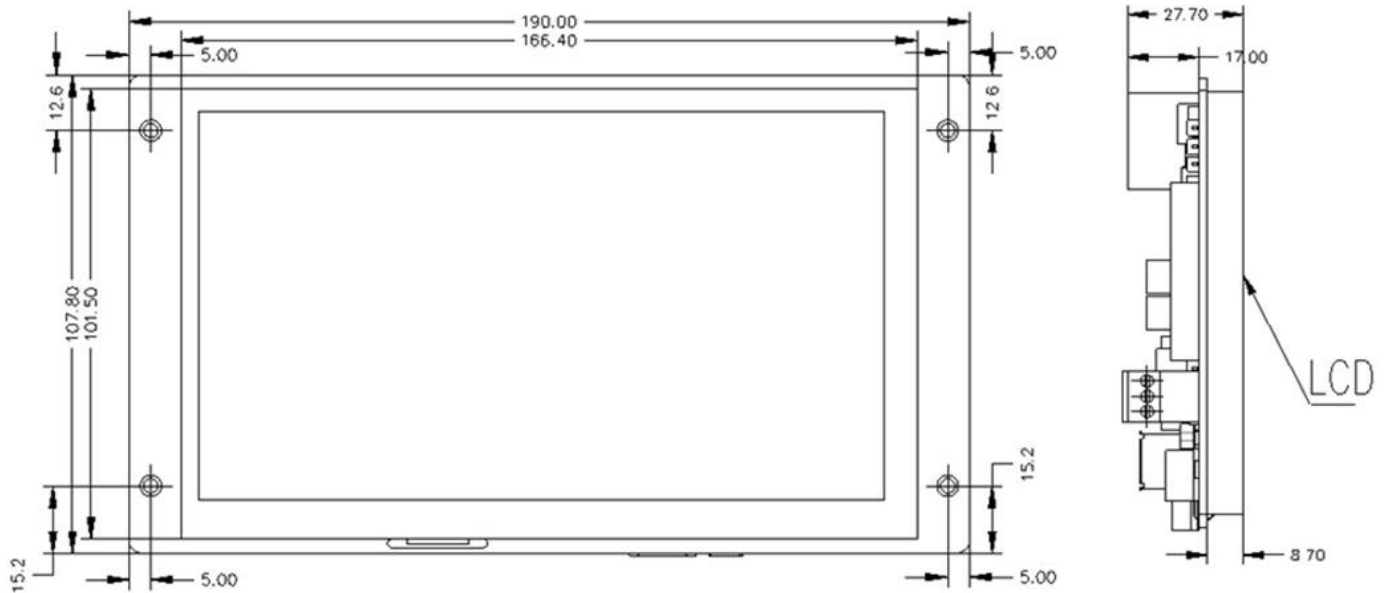
The EPC-ESP32P4-070 can be installed in an embedded enclosure ([guide](#)).

Attention

Do not apply excessive mechanical pressure to the display when mounting it in an enclosure.

Mechanical Specifications

The outer mechanical dimensions of EPC-ESP32P4-070 are 190 x 107.8 x 27.7mm (W × H × D). Please refer to the technical drawing in the figures below for details related to the specific product measurements.



Mechanical Drawing

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